

An Ethical Simulated Phishing Study in SMEs Assessing Non-Technical Employee Susceptibility and Delivering Targeted Awareness

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This study was conducted under informed participant consent in accordance with the **BPS Code of Human Research Ethics (2021)**, the **UK General Data Protection Regulation (2024)**, and the **Nigeria Data Protection Act (2023)**. No malicious activity was conducted. All data was processed, anonymised, and deleted in accordance with these legal and ethical obligations

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1. INTRODUCTION AND PROJECT AIMS

Phishing remains the leading human-exploited attack vector in global cybersecurity incidents. Verizon's 2025 Data Breach Investigations Report, analysing 22,052 incidents across 139 countries, found that 60% of confirmed data breaches involved a human element, with phishing featuring in approximately 25% of all breaches recorded (Verizon, 2025). Proofpoint's 2024 State of the Phish Report, drawing on data from 7,500 end users and 1,050 security professionals across 15 countries, found that 71% of organisations experienced at least one successful phishing attack in 2023 (Proofpoint, 2024). In SME environments, where dedicated security personnel are scarce and security cultures are often informal, individual employees bear disproportionate risk responsibility.

The challenge for SMEs is not simply technical. Firewalls, spam filters, and endpoint detection systems are routinely bypassed not through technical sophistication, but through the deliberate exploitation of human psychology. Workman (2008) demonstrated that individuals with strong computing competency were no less susceptible to social engineering than those with limited technical skills, confirming that susceptibility is a behavioural and cognitive phenomenon rather than a technical one. This study was designed with that distinction at its core.

This project was structured around three sequential phases:

Phase	Activity
Design	Infrastructure build, consent collection, simulation crafting, training module development
Execute	Campaign launch via GoPhish, participant interaction tracking, behavioural data collection via Google Form
Analyze	Data analysis, benchmark comparison, findings and recommendations

2. ETHICAL FRAMEWORK AND CONSENT

2.1 Governing Standards

The study was designed in compliance with the British Psychological Society Code of Human Research Ethics (BPS, 2021), the UK General Data Protection Regulation (2024), and the Nigeria Data Protection Act (2023). which sets out principles of scientific integrity, respect for participant autonomy, and minimisation of harm in human research contexts.

2.2 Consent Process

Prior to any study activity, all 17 participants received a plain-language consent email explaining the study's purpose, what participation would involve, and their right to withdraw at any time without consequence. Written confirmation was obtained from each participant by email before they were enrolled in the campaign.

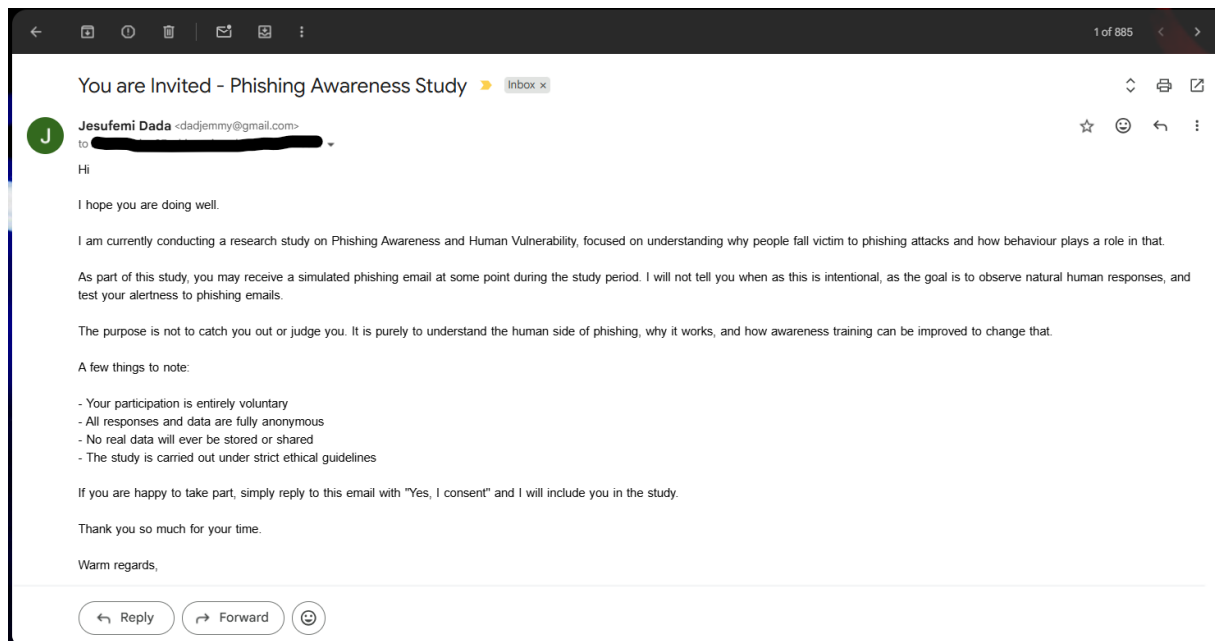
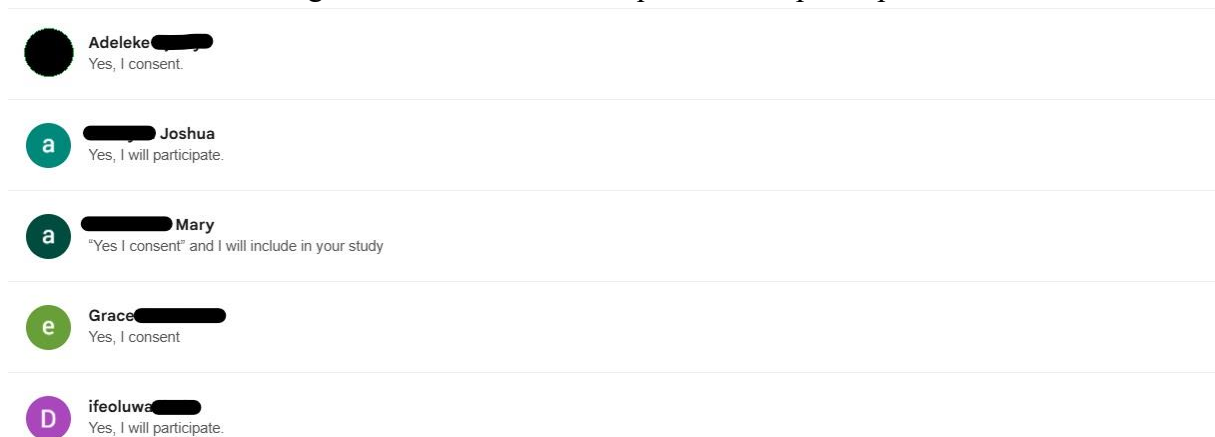


Figure 1: Consent Email request sent to participants



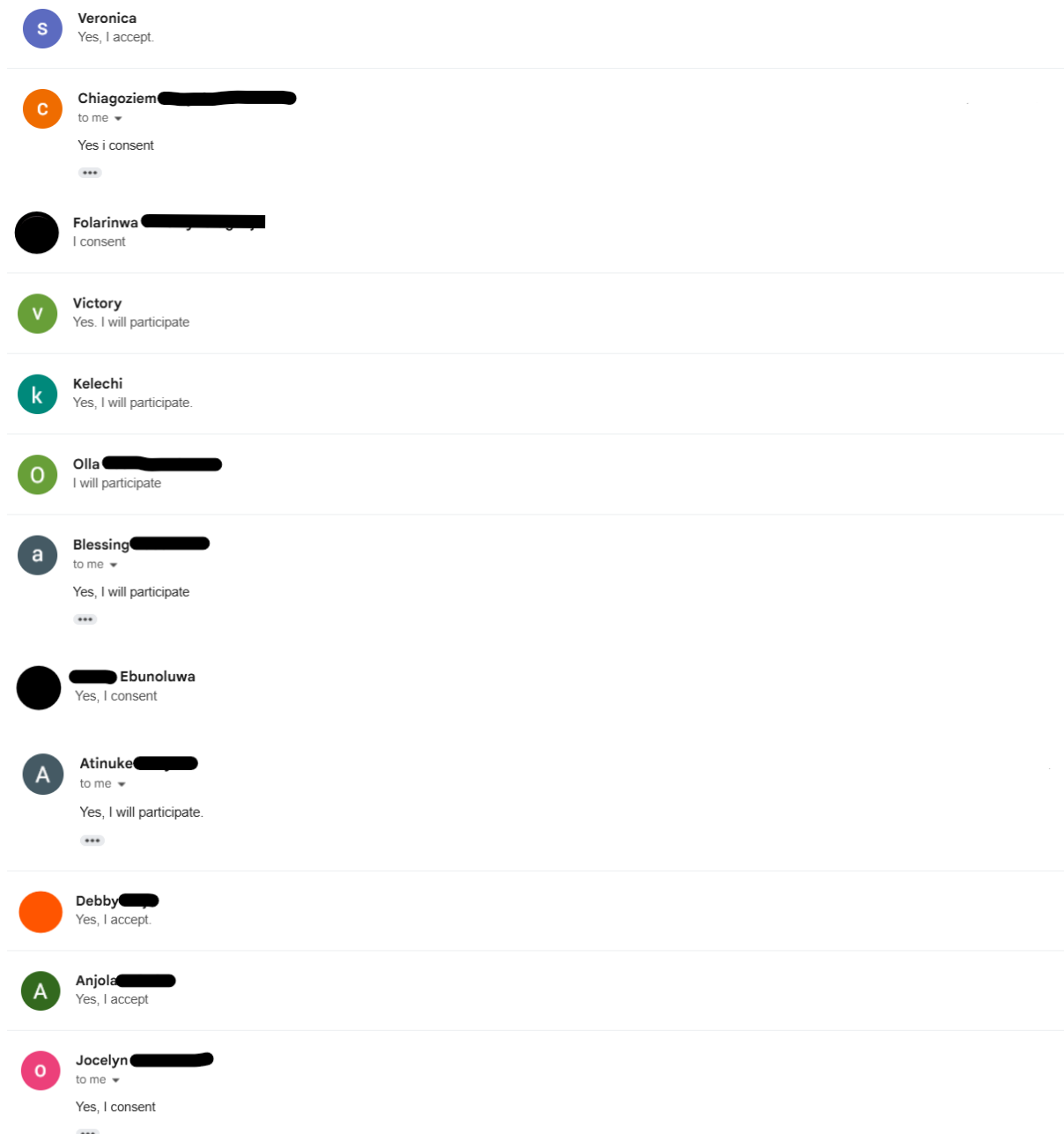


Figure 2: Screenshots of the 17 Participant email replies confirming consent.

2.3 Ethical Measures Implemented

- Informed consent obtained in advance: all 17 participants confirmed consent before receiving the simulation email.
- No organisation, brand, logo, or letterhead used: the email contained no real institutional identity, preventing impersonation of any legitimate entity (BPS, 2021).
- Email content designed to avoid distress: language was controlled to exclude panic-inducing threats or personal accusations.
- No passwords captured: the credential entry field was non-functional. No password entered by any participant was processed, stored, or transmitted.

- Email addresses encrypted in transit: any email address entered was encrypted. No plaintext personally identifiable data was stored.
- Immediate ethical disclosure on the landing page: upon interaction with the credential entry page, participants were redirected immediately to a disclosure page confirming it was the consented simulation, that their real password had not been captured, and that there was nothing to be concerned about.
- Google Form responses were fully anonymous: no names, email addresses, or identifying metadata were collected in the behavioural questionnaire.
- All raw data deleted after analysis: in compliance with the UK GDPR storage limitation and data minimisation principles (ICO, 2024).
- Right to withdraw communicated and honoured: no participant faced any consequence for non-participation.

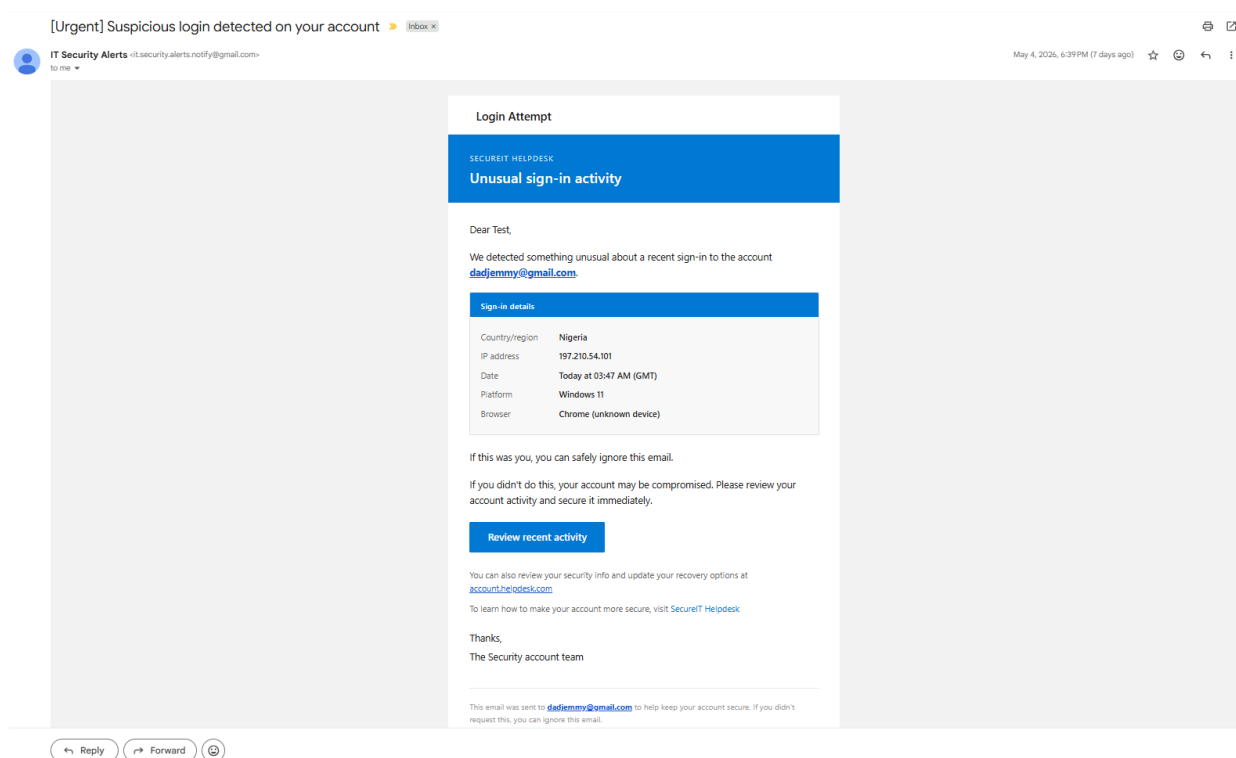


Figure 3: Phishing Email received by consented participant

This Was a Simulated Phishing Email
Phishing Awareness Training — Behavioural Study

As earlier discussed with you, which you gave consent, you just clicked a **simulated phishing link** as part of a consented phishing awareness study. **No real data was captured.** Any email you entered was automatically hashed using SHA-256, your actual password was never seen or stored by the researcher.

Before your free personalised training begins, please answer the short survey below **honestly**. Your answers are **fully anonymous** and will help understand human behaviour, not to judge you.

Privacy Notice: All responses are anonymised. No passwords were captured or stored. Data is used solely for this research study and deleted after project completion. Consent approval obtained.

Step 1 — Complete the Survey Below

Phishing Awareness — Behavioural Survey

This survey is anonymous and takes less than 2 minutes. Please answer honestly.

[Sign in to Google](#) to save your progress. [Learn more](#)

* Indicates required question

Phishing Awareness

Phishing Awareness — Behavioural Survey

Before today, how confident were you that you could spot a phishing email?

☐ Not Confident at all

☐ Slightly confident

☐ Moderately confident

☐ Quite confident

☐ Extremely confident - I thought I'd never fall for one

Submit Clear form

Google Forms This content is neither created nor endorsed by Google.

Step 2 — Start Your Free Training

Once you have submitted the survey above, click the button below to begin your personalised security awareness training. If prompted for a class code, use: **ym7zwosp**

Start My Training →

Phishing Awareness Training — Behavioural Study - Jesufemi Dada - 2026 - Consent approval obtained

Figure 4: Ethical disclosure landing page shown to participants after clicking the simulation link

2.4 GDPR Compliance

UK GDPR Principle	How Applied
Lawfulness and fairness	Prior informed consent obtained from all participants
Purpose limitation	Data collected only for stated research purposes
Data minimization	Only anonymous behavioural responses collected
Storage limitation	All raw data deleted after analysis
Integrity and confidentiality	Passwords not captured; email entries encrypted in transit

3. TECHNICAL INFRASTRUCTURE

3.1 Overview

The simulation required a cloud-hosted environment capable of sending authenticated emails, tracking participant interactions, and hosting a custom landing page, accessible from any location without a shared network dependency between researcher and participants. Four services were used, each assigned a single responsibility:

Service	Role
Namecheap	Domain registration and DNS management
DigitalOcean	Cloud VPS hosting the simulation platform
Brevo	Authenticated SMTP transactional email delivery
GoPhish	Campaign management, participant tracking, landing page, and reporting

PowerShell was used on the local Windows workstation to administer the VPS via SSH, executing setup commands, verifying service status, and managing the GoPhish process remotely.

3.2 Key Technical Outcomes

- Domain resolved correctly to the hosted DigitalOcean server.
- Brevo SMTP authentication passed sender verification checks, ensuring reliable inbox delivery.
- GoPhish was configured as a persistent system service, starting automatically on server boot.
- The study environment remained continuously available throughout the campaign without manual intervention.

Note: Full configuration details, API credentials, SMTP credentials, and operational reproduction steps are intentionally withheld from this document in the interest of responsible disclosure and unethical reproduction prevention.

4. PHISHING AWARENESS TRAINING MODULE

4.1 Design Principle

Every participant who clicked the simulation link was immediately enrolled in a custom-built phishing awareness training Module. This approach follows the just-in-time training principle, delivering education at the moment of greatest relevance. KnowBe4's 2025 benchmarking data, drawn from 67.7 million simulated phishing tests across 62,400 organisations globally, shows that continuous simulation-based training reduces the average organisational Phish-prone Percentage from 33.1% to 4.1% over 12 months (KnowBe4, 2025). The National Cyber Security Centre similarly recommends embedding training at the point of failure as a core component of organisational phishing defence (NCSC, 2025).

4.2 Delivery

The training was hosted on Google Classroom, accessed via a clearly labelled button at the bottom of the anonymous Google Form. 9 out of 12 participants who clicked the simulation link completed the training, representing a 75% training uptake rate among the susceptible cohort. Content was provided in two formats to support different learning styles and accessibility requirements: PDF for independent reading and future reference, and PDF with audio narration for auditory learners.

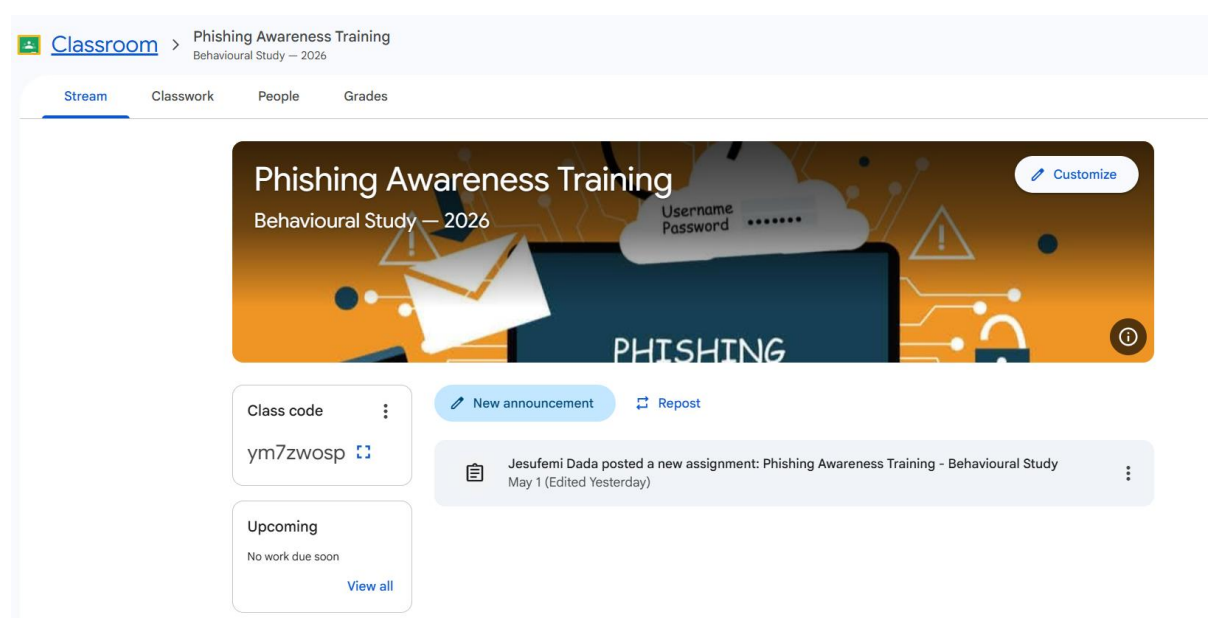


Figure 5: Google Classroom course landing page.

4.3 Module Structure

Module	Title
1	You Just Fell for a Phishing Attack: Here Is Why That Is Completely Normal
2	How to Deeply Inspect an Email Before Clicking Anything
3	What to Do If You Have Already Clicked
4	How to Investigate a Suspicious Link or IP Address
5	Your Personal 5-Second Phishing Checklist and Final Message

Module 1: reframed the simulation experience through evidence: 3.4 billion phishing emails are sent daily and phishing accounted for 36% of all data breaches globally in 2024. The module established that clicking does not reflect personal failure but reflects the precision of modern social engineering. Its opening message was: You are not stupid. You are human. And phishing attacks are specifically engineered to exploit the way human brains work under pressure.

Module 2: introduced the STOP, LOOK, VERIFY framework across five practical verification checks: verifying the actual sender email address behind the display name; understanding that inbox placement does not confirm legitimacy; hovering or long-pressing links to inspect the real destination URL before clicking; recognising emotional manipulation tactics including fear, urgency, authority, curiosity, and reward; and navigating directly to websites rather than following email links.

Module 3: provided a step-by-step incident response protocol. If no data was entered: close the tab, run a malware scan, and monitor accounts for 48 hours. If credentials were entered: immediately change the password on the real website, enable Two-Factor Authentication, check recent login activity for unrecognised sessions, check for password reuse on other accounts, and report to the NCSC at report@phishing.gov.uk.

Module 4: equipped participants with three free investigative tools: VirusTotal for scanning URLs against 70+ security engines; IPLocation.net for identifying the geographic origin and registered owner of an IP address; and WHOIS DomainTools to check domain registration age, noting that recently registered domains are a major indicator of malicious intent.

Module 5: delivered a portable five-question daily checklist and closed with the message: The fact that you clicked does not define you. The fact that you are here, learning from it, does.

URL Link to Classroom training Module:

<https://classroom.google.com/c/Nzk3NjIwMDE2OTM1?cjc=ym7zwosp>

5. METHODOLOGY

5.1 Study Design

This study used a mixed-methods cross-sectional design combining quantitative campaign data from GoPhish with ordinal and categorical self-report data from an anonymous Google Form. This approach is consistent with published phishing simulation research (Smyth, 2023). Vishwanath et al. (2011) established that an integrated information processing model incorporating both behavioural and attitudinal data is required to adequately explain phishing susceptibility, as technical data alone cannot account for the cognitive and psychological factors involved.

5.2 Participants

Seventeen participants were recruited from personal and professional networks, all employed within SME environments at the time of the study. Approximate age range: 26 to 34 years. All participation was voluntary with prior informed consent obtained from each participant.

5.3 Data Collection

GoPhish captured Three interaction states per participant: Email Sent, Email Opened, and Clicked Link with precise timestamps enabling time-to-click calculation.

The Google Form (anonymous, five questions) collected the following:

Question	Scale
Q1: Busyness at time of click	5-point ordinal
Q2: Stress at time of click	5-point ordinal
Q3: Number of concurrent tasks	4-point ordinal
Q4: Reason for clicking	Multi-select categorical
Q5: Prior confidence in detecting phishing	5-point ordinal

5.4 Analysis Approach

GoPhish data was analysed using descriptive statistics and frequency distributions, benchmarked against KnowBe4's 2025 Phishing by Industry Benchmarking Report, covering 67.7 million simulated phishing tests across 62,400 organisations globally (KnowBe4, 2025). Q4 multi-select responses were expanded so that each reason selected by any respondent was counted individually, then mapped against Cialdini's (1984) principles of influence, as applied to phishing by Ferreira, Coventry and Lenzini (2015). Google Form ordinal data was analysed using Spearman's rank correlation coefficient, the appropriate non-parametric test for ordinal data with small samples. A composite Cognitive Load Score was calculated per respondent by summing Q1, Q2, and Q3 ordinal values, with a maximum possible score of 14. All statistical results are treated as exploratory and directional, appropriate to the pilot-scale sample size of $n=17$.

6. DATA ANALYSIS AND FINDINGS

6.1 Campaign Performance

GoPhish recorded the following interaction outcomes across all 18 participants:

Status	Count	Percentage of 18 Sent
Clicked Link	12	70.6%
Opened, no click	4	23.5%
Sent, no open	1	5.9%

Of the 12 who clicked, 11 (91.7%) completed the anonymous Google Form, and 9 out of 11 (75%) completed the Google Classroom training programme.

Benchmark comparison: KnowBe4's 2025 Phishing by Industry Benchmarking Report found a global average baseline Phish-prone Percentage of 33.1% across all organisation sizes and 24.6% for organisations of 1 to 250 employees, the most directly comparable SME band (KnowBe4, 2025). The 70.6% click rate observed in this study is 2.9 times the SME-specific industry baseline, confirming the cohort was substantially more susceptible than average prior to any awareness training.

6.2 Why Participants Clicked

Q4 was a multi-select question, allowing respondents to choose more than one reason. The dataset records 18 total reason mentions across 11 respondents. Results below represent the number of respondents who selected each option:

Click Reason	Respondents	% of 11	Cialdini Principle
I was curious about what it said	6	54.5%	Scarcity / Curiosity
I thought the email was genuine	5	45.5%	Authority
I was in a rush and did not stop to think	4	36.4%	Distraction
The email looked official or professional	2	18.2%	Authority
I clicked automatically without really thinking	1	9.1%	Automaticity
I panicked or felt worried by the message	0	0.0%	None

Curiosity was the most frequently reported driver, cited by 6 of 11 respondents (54.5%). This aligns with Cialdini's (1984) scarcity principle, the suggestion of exclusive or personally relevant information that motivates engagement without deliberate evaluation. Ferreira, Coventry and Lenzini (2015) confirmed through analysis of phishing emails that curiosity and

authority are among the most commonly deployed persuasion techniques in phishing campaigns.

Three respondents selected three or four reasons simultaneously, indicating compound susceptibility: curiosity, genuine belief in the email, rushing, and in one case automatic clicking all converged at once. These respondents were also among the most cognitively loaded in the dataset (see Section 6.3).

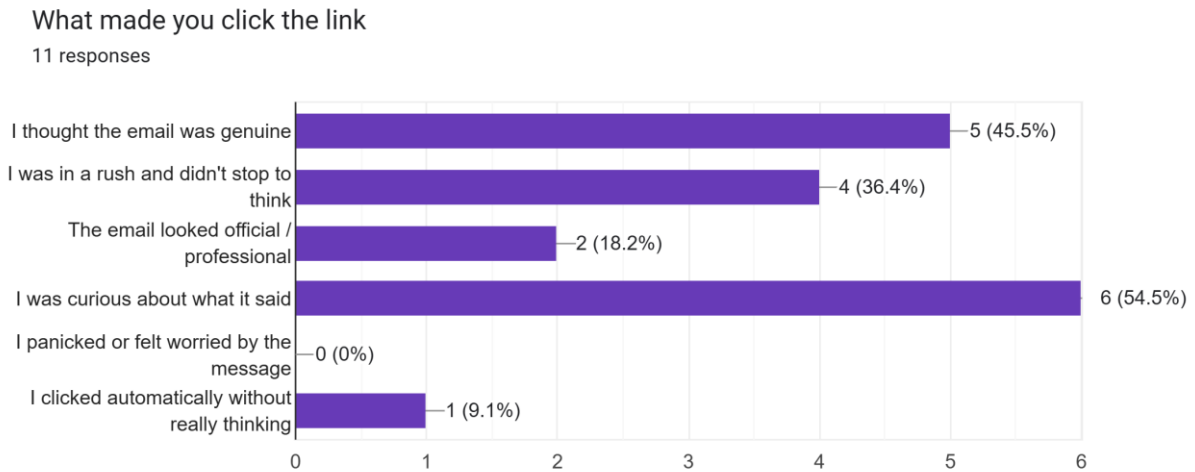


Figure 6: Self-Reported Reasons for Clicking (Q4, Multi-Select, n=11)

6.3 Cognitive Load at the Time of Click

A composite Cognitive Load Score was calculated per respondent (Q1 + Q2 + Q3; maximum = 14):

Statistic	Value
Mean	8.82 / 14
Median	7.00
Standard Deviation	2.86
Range	5 to 14

Busyness distribution (Q1):

Level	Count
Moderately busy	4
Slightly busy	3
Very busy, juggling multiple things	2
Extremely busy or Overwhelmed	2

Stress distribution (Q2):

Level	Count
Very calm and relaxed	3
Quite stressed	3
Slightly stressed	2
Very stressed	2
Moderately stressed	1

Concurrent tasks distribution (Q3):

Level	Count
1 to 2 other tasks	5
3 to 5 other tasks	5
More than 5 tasks	1

The stress profile shows a notably polarised distribution: 5 respondents were Quite Stressed or Very Stressed at the time of clicking, while 3 were Very Calm and Relaxed. This bimodal pattern is analytically significant. Susceptibility occurred at both ends of the stress spectrum, indicating the phishing email was effective regardless of whether participants were calm or under pressure. Vishwanath et al. (2011) found that urgency cues in phishing emails stimulate increased information processing, paradoxically reducing the cognitive resources available to detect deception, which supports why susceptibility can occur even in low-stress conditions when curiosity is the operative trigger. The mean Cognitive Load Score of 8.82 indicates a moderate to high load environment across the cohort.

The Spearman correlation between Cognitive Load and prior confidence was $\rho = 0.461$ ($p = 0.154$), and between Stress and Confidence was $\rho = 0.418$ ($p = 0.201$). Both indicate moderate positive directional trends but do not reach conventional significance thresholds, consistent with the exploratory nature of this pilot study.

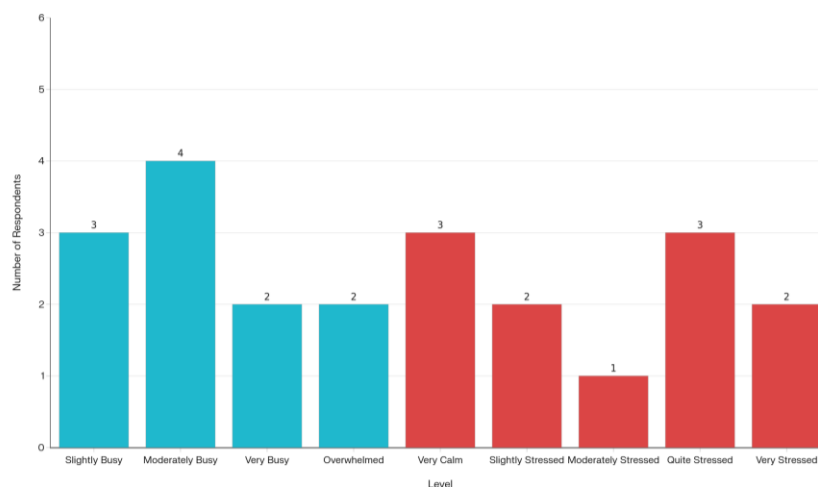


Figure 7: Busyness and Stress at Time of Click (n=11)

6.4 Time-to-Click Behaviour

GoPhish timestamps recorded the precise moment each participant clicked relative to email delivery:

Category	Range	Count	Percentage of Clickers
Impulsive	Under 5 minutes	3	25.0%
Fast	5 to 30 minutes	1	8.3%
Delayed	30 minutes to 2 hours	5	41.7%
Late	Over 2 hours	3	25.0%

Mean time-to-click: 76.1 minutes (Standard Deviation = 83.9 minutes; Range = 0.05 to 253.1 minutes)

75% of clicks occurred more than 30 minutes after delivery. Only three participants clicked within the first five minutes. The high standard deviation (83.9 minutes) reflects the wide range of engagement patterns across the cohort. The majority of participants had time to deliberate and still clicked, confirming that the email was sufficiently convincing to survive reflection. This reinforces the conclusion from Q4 that susceptibility was driven by the quality of social engineering rather than momentary inattention.

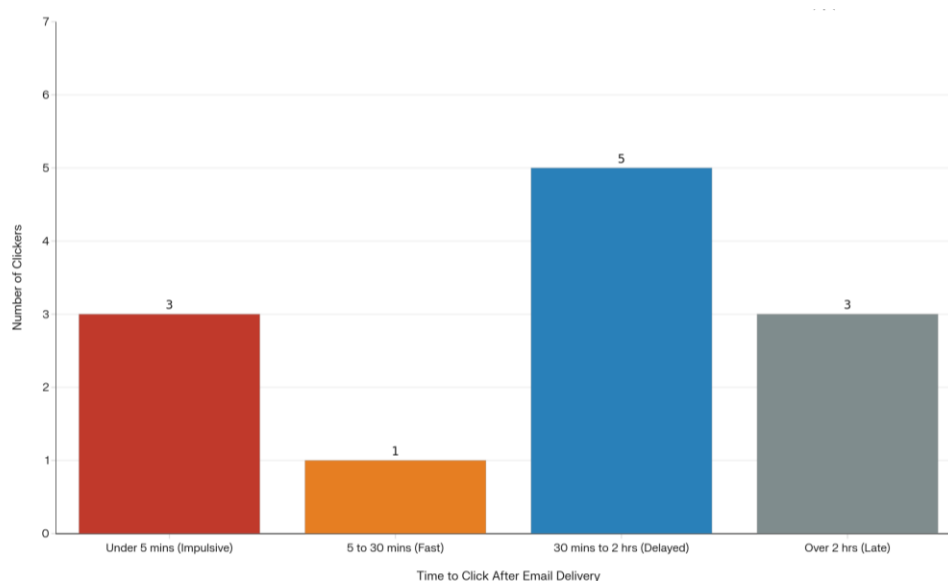


Figure 8: Time-to-Click Distribution (n=12)

6.5 The Overconfidence Paradox

Q5 asked participants to rate their confidence in spotting a phishing email before the study:

Confidence Level	Count	Percentage
Slightly confident	2	18.2%
Moderately confident	3	27.3%
Quite confident	5	45.5%
Extremely confident, I thought I would never fall for one	1	9.1%

6 of 11 respondents (54.5%) rated themselves as Quite Confident or Extremely Confident before the study, and every one of them clicked the link. Examining those six respondents' Q4 responses:

- One respondent (Extremely Confident) cited rushing and not stopping to think.
- Two respondents (Quite Confident) cited curiosity, genuine belief, and rushing simultaneously.
- One respondent (Quite Confident) cited curiosity as the sole reason.
- One respondent (Quite Confident) cited genuine belief in the email.
- One respondent (Quite Confident) selected four simultaneous reasons including automatic clicking.

This pattern is directly consistent with the Dunning-Kruger effect: individuals overestimate their competence in a domain because they lack sufficient knowledge to recognise the limits of what they do not yet know (Kruger and Dunning, 1999). In the security context, this phenomenon is well established. Workman (2008) demonstrated that compliance with social engineering was significantly predicted by social proof and authority cues regardless of the individual's stated technical confidence. Employees who have had minimal formal security training commonly report elevated phishing detection confidence precisely because they have not encountered a well-crafted simulation or live attack (Kruger and Dunning, 1999). A training programme targeting only self-identified low-confidence individuals will therefore systematically fail to reach the majority of the vulnerable population identified in this study.

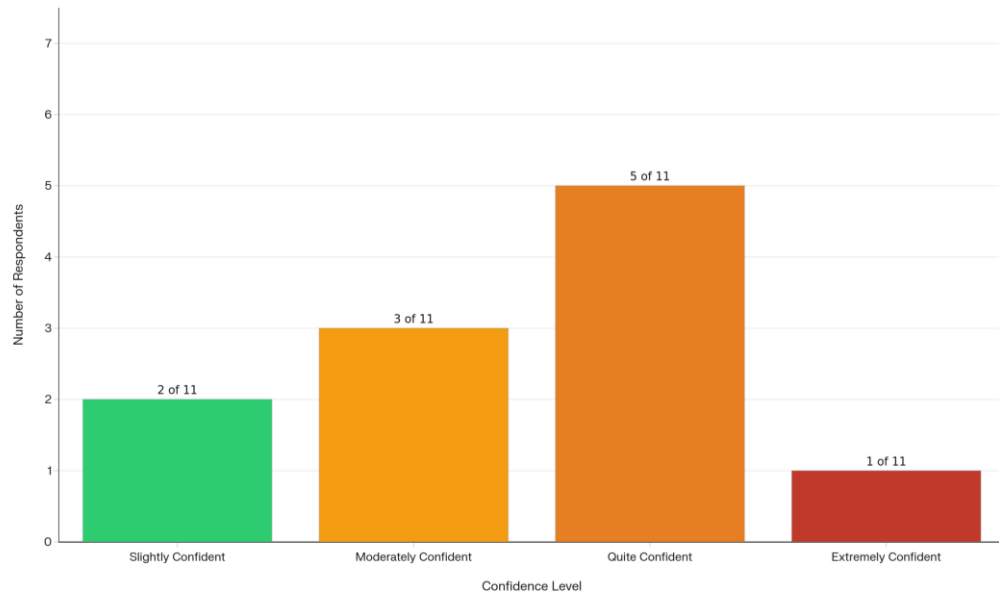


Figure 9: Prior Confidence in Phishing Detection (n=11)

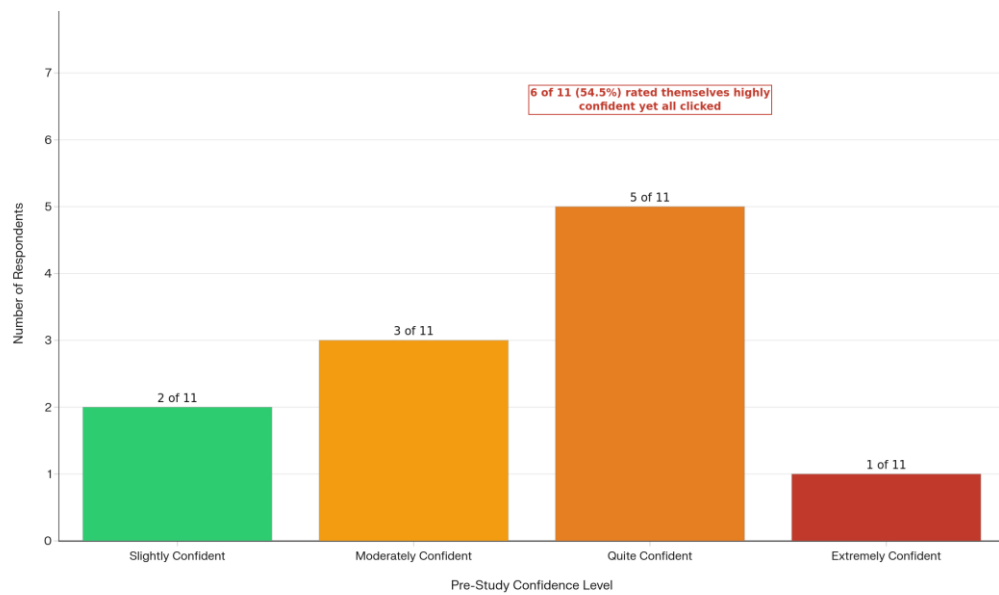


Figure 10: The Overconfidence Paradox (n=11)

6.6 Statistical Summary

Test	Variables	Result	Interpretation
Spearman's rho	Confidence vs Cognitive Load	$\rho = 0.461, p = 0.154$	Moderate positive trend, not significant
Spearman's rho	Confidence vs Stress	$\rho = 0.418, p = 0.201$	Moderate positive trend, not significant
Spearman's rho	Busyness vs Tasks	$\rho = 0.480, p = 0.135$	Moderate positive trend, not significant
Descriptive	Time to click	Mean = 76.1 min, SD = 83.9	High variability; majority clicked after deliberation
Descriptive	Cognitive Load Score	Mean = 8.82 / 14, SD = 2.86	Moderate to high load across cohort

The moderate positive Spearman values suggest a directional pattern worth investigating in larger samples: higher self-confidence may be associated with higher cognitive load states, potentially because confident individuals take on more concurrent tasks and are more cognitively occupied at the moment of clicking. These results are exploratory and should be interpreted accordingly.

7. CONCLUSIONS

Susceptibility significantly exceeded industry norms. The 70.6% click rate was 2.9 times KnowBe4's (2025) SME baseline of 24.6% for organisations of 1 to 250 employees, confirming that an untrained SME-employed cohort represents a high-risk population before any security awareness intervention is in place.

Curiosity and deception, not panic, drove susceptibility. The Q4 data identifies curiosity (54.5%) and genuine belief in the email (45.5%) as the dominant click motivators.

Overconfidence is the most critical and underaddressed vulnerability. 54.5% of form respondents believed they were well-equipped to detect phishing before the study, yet all clicked. The Dunning-Kruger pattern (Kruger and Dunning, 1999) means that standard awareness programmes targeting self-identified low-confidence employees will systematically miss the majority of the at-risk population. Confidence calibration must be a central design principle in any SME training programme.

8. RECOMMENDATIONS

- 1. Run recurring phishing simulations with embedded just-in-time training.** KnowBe4 (2025) data shows continuous simulation reduces average organisational PPP from 33.1% to 4.1% over 12 months. Single-event simulations produce only temporary behavioural change. Minimum frequency for SMEs should be quarterly, with immediate embedded training triggered on each click.
- 2. Target overconfident employees explicitly.** Over half of susceptible respondents were self-rated as highly confident. Use confidence calibration exercises, asking employees to predict whether they would click before a simulation then comparing against actual results, to address the metacognitive gap identified by Kruger and Dunning (1999).
- 3. Build training around curiosity and authority cues, not just fear.** Curiosity and email legitimacy drove the majority of clicks. Training must address display name spoofing, look-alike domain construction, the mechanics of curiosity-triggering subject lines, and the STOP, LOOK, VERIFY framework from Module 2. Ferreira, Coventry and Lenzi (2015) confirm that authority and curiosity are the primary persuasion vectors in effective phishing.
- 4. Design for habitual vigilance across all stress states.** Participants clicked under both calm and high-stress conditions. Training should develop procedural habits that operate automatically regardless of cognitive state: hover before clicking, verify the sender address against the display name, and navigate directly to websites rather than following embedded links (NCSC, 2025).
- 5. Establish a clear phishing reporting pathway.** No respondent cited reporting as a response behaviour. Every SME should communicate the NCSC reporting address (report@phishing.gov.uk) to all staff and integrate it into onboarding and refresher training materials.
- 6. Maintain accessible dual-format training content.** The PDF and PDF with audio narration delivery model used in this study accommodates different learning styles and accessibility requirements. Accessible training formats should be treated as a baseline requirement rather than an optional addition.

9. LIMITATIONS

Limitation	Note
Small sample (n=17)	Results are exploratory; not statistically generalisable to the wider population.
Convenience sampling	Social familiarity with the researcher may have elevated baseline trust in the email, potentially inflating click rates.
Single phishing template	Results reflect susceptibility to one specific social engineering design. Susceptibility rates may differ with other templates.
No pre/post knowledge test	Training impact on knowledge cannot be measured quantitatively from this dataset.
No longitudinal follow-up	Whether behavioural change was durable after training was not assessed.
Self-report bias (Q4)	Click reasons reflect post-hoc recall. Respondents may rationalise rather than accurately reconstruct their in-the-moment motivation.

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